

Clinical Science

Long-term survival after ruthenium plaque radiotherapy for uveal melanoma. A meta-analysis of studies including 1,066 patients

Stefan Seregard

Ophthalmic Pathology and Oncology Service, St Eriks Eye Hospital and Karolinska Institutet, Stockholm, Sweden

ABSTRACT.

Purpose: Most case series of patients managed by brachytherapy for uveal melanoma are small and survival data show considerable inter-study variation. The aim of this study was to summarise similarly structured case series by meta-analysis.

Methods: A systematic review generated 5 similarly structured case series including survival data for 1,066 patients treated by ruthenium plaque radiotherapy for uveal melanoma. After assessing the inter-study clinical heterogeneity, data were weighed for study size and pooled.

Results: Patient and radiotherapy characteristics were largely homogenous, but tumour size varied considerably between studies. The 5-year melanoma-related mortality rate was 6% for small and medium tumours (T1/T2) and 26% for large (T3) tumours. The 5-year and 10-year melanoma-related mortality rates for a balanced set of tumours with small, medium and large tumours being present in similar proportions were 14% and 22%, respectively.

Conclusion: Case series of patients with uveal melanoma managed by brachytherapy may be pooled to increase sample size and study power. The present estimate of survival following ruthenium plaque radiotherapy compares favourably with previously summarised data of survival after enucleation for similarly sized tumours.

Key words: Uveal melanoma – ruthenium plaque radiotherapy – survival – meta-analysis

Acta Ophthalmol. Scand. 1999; 77: 414–417

Copyright © Acta Ophthalmol Scand 1999. ISSN 1395-3907

Uveal melanoma is the most common primary intraocular malignancy. Enucleation used to be the standard treatment, but there is a current trend to manage this tumour by any of a range of eye-preserving options, including charged particle irradiation, plaque radiotherapy and local resection (Seregard 1995). During the last decades, plaque radiotherapy has become increasingly popular. It involves fixing a disc-shaped radiation source on the sclera covering the tumour, and commonly used radioactive sources include cobalt-60,

iodine-125, palladium-103 and ruthenium-106. Different low-energy sources are now in common use, and iodine-125 and ruthenium-106 are the most common used nuclides in North America and Europe, respectively (Finger 1997).

Results from randomised trials of uveal melanoma are scarce and many studies of plaque radiotherapy include tumours managed by more than one type of radioactive source. Because the relative dose distributions are distinctly different for each nuclide used for plaque radiotherapy, the selection of a particular nuclide may

influence local control rate and hence the metastatic rate (Finger 1997). In addition, most case series of patients treated with a single nuclide are limited to 100–300 patients, usually reflecting the experience of a single centre. Such data tend to show considerable inter-study variation, but data using the same end-points may be pooled for meta-analysis. Though meta-analysis is most often used to synthesize the results of clinical trials it may also summarise the rate of occurrence of an event, like tumour-related death. Meta-analysis has been used to assess survival after enucleation for uveal melanoma (Diener-West et al. 1992), but has not been applied to studies based on an eye-preserving therapy. The aim of this study was to provide a systematic review of survival data obtained after ruthenium plaque radiotherapy and to pool comparable studies for a better prognostic estimate.

Material and Methods

Ruthenium-106 remains a commonly used nuclide for brachytherapy of uveal melanoma since the introduction more than 30 years ago and the standard radiotherapy dose has remained essentially unchanged over time. Based on this, the following criteria for including a previous study in the current overview were adopted. Ruthenium-106 had been used as the sole radiation source for episcleral brachytherapy of uveal melanoma with the treatment aimed to deliver 80–100 Gy at the tumour apex. Each centre should have reported 5-year survival rates (defined as all-cause or cause-specific) for a minimum of 50 patients. When two or more studies appar-

Table 1. Included case series.

Centre	Author(s)	Time period	Patients	F-up (mean)	F-up (range)
Leipzig	Lommatzsch	1964–1984	309	6.7 yrs	1.0–21.0 yrs
Hamburg	Kleineidam et al.	1975–1986	184	6.1 yrs	0.1–13.3 yrs
Helsinki	Summamanen et al.	1981–1991	100	3.3 yrs	1.1–10.1 yrs
Lyon	Grange et al.	1983–1993	207		0.1–10.0 yrs
Stockholm	Seregard et al.	1979–1995	266	3.6 yrs	0.5–12.5 yrs

ently included identical patients only the largest study was accepted.

A systematic selection of all studies including survival after ruthenium plaque radiotherapy was then performed using the MEDLINE database accessed at the National Library of Medicine website at the Internet (<http://WWW.ncbi.nlm.nih.gov/PubMed/medline.html>) on April 18, 1998. The combination of the searchwords “ruthenium and melanoma and (uvea or choroid)” generated 87 articles which were reviewed by title. Based on this title review, 23 papers were selected for abstract review and 16 studies including survival data were chosen for complete review. Five of these studies met all inclusion criteria and were accepted for meta-analysis; the pooled number of patients being 1,066 (Grange et al. 1995; Kleineidam et al. 1993; Lommatzsch 1986; Seregard et al. 1997; Summanen et al. 1993). Each study covered the experience of a single centre over a period of 10 years or more (Table 1).

To assess if the data to be summarised were homogenous, patient, tumour and radiotherapy characteristics were subtracted from each of the included studies. Uveal melanoma is believed to be slightly more common in men (Jensen 1970), but in all but one of the included case series women were overrepresented by 20–40% (Table 2). The age structure of the respective study cohort was reasonably similar with minimal deviation of age range and mean or median age (Table 2). Similarly, follow-up periods were largely homogenous with regard to range and mean or median follow-up time (Table 1).

Table 2. Patient characteristics.

Centre	Male:female ratio	Median age (yrs)
Leipzig	1:0.84	52
Hamburg	1:1.27	58
Helsinki	1:1.43	59
Lyon	1:1.25	60
Stockholm	1:1.27	63

Only 3 of the included studies contained data on tumour location; in 2 of these studies the proportion of tumours anterior of the equator was almost identical, but the third case series contained a high proportion (38%) of anterior melanomas (Table 3). The size of tumours varied extensively between the included case series. This was evident from the proportions of tumours classified as small (T1), medium (T2) or large T3) and from the proportions of tumours more than 10 mm in largest diameter (Table 3). Because biopsy specimens for histopathological examination were not available for any case series, tumour cell type

Table 3. Tumour characteristics.

Centre	T1 (%)	T2 (%)	T3 (%)	LTD>10 mm (%)	Anterior (%)
Leipzig	71	8	21	29	
Hamburg	45	40	15	55	
Helsinki	7	37	56	93	38
Lyon	6	40	53	91	20
Stockholm	15	48	37	85	21

LTD=largest tumour diameter

Table 4. Brachytherapy characteristics.

Centre	Nuclide	Tumour apex dose	Scleral dose
Leipzig	Ru-106	>100 Gy	<1500 Gy
Hamburg	Ru-106	median=250 Gy	median=747 Gy
Helsinki	Ru-106	median=100 Gy	<1200 Gy
Lyon	Ru-106	>60 Gy	<1500 Gy
Stockholm	Ru-106	median=100 Gy	median=543 Gy

Table 5. Melanoma-related and all-cause (within brackets) mortality after ruthenium plaque radiotherapy.

Centre	Method estimate	5-yr mortality (%)	10-yr mortality (%)	5-yr mortality (%), T1/T2 vs T3
Leipzig	life table	11 (16)	20 (34)	
Hamburg	Kaplan-Meier	(16)	(27)	
Helsinki	cumulative	22	34	3 vs 39
Lyon	cumulative	17 (20)	19 (29)	9 vs 24
Stockholm	Kaplan-Meier	14	24	5 vs 21

could not be assessed. Whereas radiotherapy data were largely homogenous (Table 4), a small number of tumours received very low radiation doses, but there was no evidence of significant inter-study variability. Based on the review of patient, tumour and radiotherapy characteristics, it was decided to present survival data subgrouped for tumour size when possible. Data were not subgrouped for gender or anterior tumour margin, because it is currently believed that the prognostic significance of these features is low (McLean 1996).

Four of the 5 included studies reported cause-specific survival based on melanoma-related deaths, 3 of 5 of these studies reported both cause-specific survival and all-cause survival and one study included only all-cause survival data (Table 5). Because survival data based on melanoma-related deaths only, usually provides more meaningful information for the clinician than all-cause survival, it was decided to base the meta-analysis on cause-specific survival. To match data from other case series, we adjusted the

survival data from the sole study reporting all-cause survival only using a similar approach as reported previously (Diener-West et al. 1992). Briefly, the relationship obtained from 2 case series providing both cause-specific and all-cause survival rates indicated that reducing all-cause survival by one-third would provide a reasonable estimate of survival adjusted for melanoma-related deaths only (Grange et al. 1995; Lommatzsch 1986). After checking that the age structure of the pooled case series was similar or nearly similar (Table 2), this correction factor was then used to adjust the all-cause survival rates obtained from one of the studies (Kleineidam et al. 1993). All pooled data were then weighed to adjust for the difference in study size.

Results

All studies included 5-year and 10-year survival rates. Because tumour size varied extensively between case series, survival data correlated to tumour size

were sought. Three studies contained such subgrouped survival data based on melanoma-related deaths only (Grange et al. 1995; Seregard et al. 1997; Summanen et al. 1993). In these studies, the 5-year mortality rate) increased 3–10 times for patients with small/medium tumours compared to patients with large tumours, indicating a strong and consistent correlation of prognosis to size (Table 5). After pooling the patients of these 3 studies (n=573), the 5-year weighed melanoma-related mortality rates were 6% and 26%, for small/medium (T1+T2) tumours and large (T3) tumours respectively, suggesting a more than fourfold increase in tumour death for patients with large tumours (Table 6).

The summarised data for all patients of all studies (n=1,066) indicated that the overall, weighed, melanoma-related 5-year and 10-year mortality rates were 14% and 22%, respectively. The proportions of small (T1), medium (T2) and large (T3) tumours in this pooled cohort (n=1,066) were almost identical or 33.9%, 32.6% and 33.5%, respectively (Table 6).

Most case series included an assessment of potential prognostic indicators by univariate or multivariate analysis. Whereas the results of a few studies suggested that older patients were more likely to die of metastatic melanoma, the most consistent finding was a statistically significant association of tumour size with prognosis (Tables 7–8).

Discussion

Our analysis shows that 5 previous studies were designed with sufficient consistency and contained similarly structured data to allow pooling of survival data. The total number of patients included in these studies (n=1,066) widely exceeds that of any single case series, permitting a more reliable estimate of patient outcome after plaque radiotherapy for uveal melanoma.

The comparatively small number of patients included in most studies of plaque radiotherapy for uveal melanoma reduces statistical power. Also, the experience of a single center is likely to be biased with regard to the selection of patients and tumours for conservative therapy. Our data suggest that pooling patients from similarly structured case series using common end-points reduces bias and may be used as a cost-effective

alternative to embark on a new larger study. However, potential sources of heterogeneity among pooled studies need to be recognised. In the present meta-analysis, the main source of clinical heterogeneity was the inter-study difference in size of tumours selected for plaque radiotherapy. While only 29% of tumours managed by brachytherapy at one centre were larger than 10 mm in diameter (Lommatzsch 1986), more than 90% of patients treated at 2 other centres had large tumours. By comparing only data that could be subgrouped for small/medium or large tumours, the inter-study agreement improved and most of the remaining difference could be attributed to random variation (Table 5).

Our results may be compared with results obtained from a meta-analysis of survival after enucleation for uveal melanoma (Diener-West et al. 1992). In this report, the 5-year all cause mortality rates were 16% for small tumours, 32% for medium tumours and 53% for large tumours. The authors adjusted melanoma-related mortality to estimate the all-cause survival by (when required) conservatively inflating the melanoma-related rates by 6% (based on earlier reports indicating a difference of 6–26% between the rates). A backward adjustment of their data generates 5-year melanoma-related mortality estimates of 15%, 30% and 50%, for patients with small, medium and large tumours, respectively. These rates are substantially higher than our 5-year melanoma-related mortality estimate of 6% for patients treated with ruthenium plaque radiotherapy for small/medium uveal melanoma and 26% for patients treated for large tumours. Similarly, the pooled, 5-year, all-cause mortality rates after enucleation appears to be higher than the all-cause mortality after plaque radiotherapy available in our summarised case series even if based on inflating melanoma-related rates by as much as 26% (e.g. 53% vs 37% for large tumours). However, significant, unrecognised bias may be present and caution is advised when comparing survival rates based on non-randomised data.

References

- Diener-West M, Hawkins B, Markowitz J A & Schachat A P (1992): A review of mortality from choroidal melanoma. II. A meta-analysis of 5-year mortality rates following enucleation, 1966 through 1988. *Arch Ophthalmol* 110: 245–250.

Table 6. Weighed melanoma-related mortality after ruthenium plaque radiotherapy.

Patients	Size	5-yr mortality (%)	10-year mortality (%)
308	T1/T2	6	
265	T3	26	
1066	all	14	22

Table 7. Prognostic indicators after ruthenium plaque radiotherapy (univariate analysis).

Centre	Size	Site	Age	Sex
Leipzig				
Hamburg	yes			
Helsinki	yes	yes		
Lyon	yes			
Stockholm	yes	no	yes	no

Table 8. Prognostic indicators after ruthenium plaque radiotherapy (multivariate analysis).

Centre	Size	Site	Age	Sex
Leipzig				
Hamburg	yes		yes	no
Helsinki				
Lyon				
Stockholm	yes	no	yes	no

- Finger PT (1997): Radiation therapy for choroidal melanoma. *Surv Ophthalmol* 42: 215–232.
- Grange J D, Joshi G, Gérard J P, Jean-Louis B, Senténac I & Ayzac L (1995): Bilan de dix années d'expérience de la bêta-curiothérapie de contact par applicateur de ruthénium 106 pour mélanome de l'uvée. *Revue de 207 cas et hommage à P.K. Lommatzsch. Ophthalmologie* 9: 317–323.
- Jensen OA (1970): Malignant melanomas of the human uvea. Recent follow-up of cases in Denmark 1943–1952. *Acta Ophthalmol (Copenh)* 48: 1113–1128.
- Kleineidam M, Guthoff R & Bentzen S (1993): Rates of local control, metastasis, and overall survival in patients with posterior uveal melanomas treated with ruthenium-106 plaques. *Radiother Oncol* 28: 148–156.
- Lommatzsch PK (1986): Results after β -irradiation ($^{106}\text{Ru}/^{106}\text{Rh}$) of choroidal melanomas: 20 years' experience. *Br J Ophthalmol* 70: 844–851.
- McLean IW (1996): Uveal nevi and malignant melanomas. In: Spencer WH (ed). *Ophthalmic Pathology. An Atlas Textbook*. Volume 3, p. 2163. WB Saunders, Philadelphia, PA.
- Seregard S (1995): Posterior uveal melanoma. The Swedish perspective. *Acta Ophthalmol Scand* 74: 315–329.
- Seregard S, af Trampe E, Lax I, Kock E & Lundell G (1997): Results following episcleral ruthenium plaque radiotherapy for posterior uveal melanoma. *Acta Ophthalmol Scand* 75: 11–16.
- Summanen P, Immonen I, Heikkonen J, Tommila P, Laatikainen L & Tarkkanen A (1993): Survival of patients and metastatic and local recurrent growth in malignant melanoma of the uvea after ruthenium plaque radiotherapy. *Ophthalmic Surg* 24: 82–90.

Received on October 26th, 1998.
Accepted on January 21st, 1999

Corresponding author:

Stefan Seregard, MD
Ophthalmic Pathology and Oncology Service
St Eriks Eye Hospital
Polhemsgatan 50
SE-112 82 Stockholm
Sweden